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RESEARCH ARTICLE

Detecting Machine-Generated Passages in Russian-Language Scientific Abstracts

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ABSTRACT

The widespread adoption of large language models has confronted scholarly journals with the task of identifying texts created wholly or partly by automated means. Most existing detectors were developed for English, and their applicability to Russian-language scientific texts is poorly understood. This article presents a new annotated corpus of 12,000 abstracts of scientific articles in Russian, half of which were written by humans and half of which contain passages generated by three different language models under different prompts. Four classes of methods are compared on this corpus: statistical perplexity features, stylometric features, a fine-tuned transformer-based classifier and a general-purpose commercial detector. The fine-tuned classifier performed best (document-level F1 = 0.91), but its performance fell to 0.74 when tested on texts generated by a model not seen during training. At the sentence level, all methods perform considerably worse, and the false-positive rate on abstracts written by authors for whom Russian is not a first language reaches 11%. These results indicate that automated detectors cannot serve as the sole basis for editorial decisions. The corpus and code are openly available for further research.

Keywords: large language models; machine-generated text detection; scientific abstracts; Russian language; research integrity

Распознавание машинно сгенерированных фрагментов в русскоязычных научных аннотациях

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АННОТАЦИЯ

Широкое распространение больших языковых моделей поставило перед научными журналами задачу выявления текстов, полностью или частично созданных автоматически. Большинство существующих детекторов разработаны для английского языка, и их применимость к русскоязычным научным текстам изучена слабо. В статье представлен новый размеченный корпус из 12 тыс. аннотаций научных статей на русском языке, в котором половина текстов написана людьми, а половина содержит фрагменты, сгенерированные тремя различными языковыми моделями с разными инструкциями. На этом корпусе сравниваются четыре класса методов: статистические признаки перплексии, стилометрические признаки, дообученный классификатор на основе трансформера и коммерческий детектор общего назначения. Наилучшие результаты показал дообученный классификатор (F1 = 0,91 на уровне документа), однако его качество снижалось до 0,74 при тестировании на текстах, сгенерированных моделью, не встречавшейся при обучении. На уровне отдельных предложений все методы работают существенно хуже, а доля ложных срабатываний на аннотациях, написанных авторами, для которых русский язык не является родным, достигает 11 %. Эти результаты свидетельствуют о том, что автоматические детекторы не могут служить единственным основанием для редакционных решений. Корпус и код опубликованы в открытом доступе для дальнейших исследований.

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Ключевые слова: большие языковые модели; детекция сгенерированного текста; научные аннотации; русский язык; академическая этика

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1 INTRODUCTION

Since 2023, editors of scholarly journals have increasingly encountered manuscripts in which parts of the text appear to have been produced by large language models. While the use of such tools for language editing is often acceptable, undisclosed generation of substantive content raises concerns about research integrity. Many publishers have adopted policies requiring disclosure, and automated detectors are being marketed as a means of enforcement.

These detectors, however, have been developed and evaluated primarily on English text. Russian, with its rich morphology and distinctive academic register, presents a different challenge. The aims of this study are to build a realistic benchmark of Russian-language scientific abstracts and to evaluate how well current detection methods perform on it, with particular attention to generalisation and fairness.

2 MATERIALS AND METHODS

Human-written abstracts (n = 6,000) were sampled from open-access Russian journals published before 2021, stratified across eight broad disciplines. For the generated half, three language models were prompted to write a full abstract from a title and keywords, to continue a human-written opening sentence, or to paraphrase a human-written abstract. Each generated text was checked by an annotator for fluency, and sentence-level labels were recorded for mixed texts.

We compared (i) a logistic regression on perplexity and burstiness features computed with an open Russian language model, (ii) a gradient-boosted classifier on 140 stylometric features, (iii) a multilingual transformer encoder fine-tuned on the training split and (iv) a commercial general-purpose detector. A leave-one-generator-out protocol was used to assess generalisation.

3 RESULTS

Document-level results are summarised in Table 1. The fine-tuned classifier achieved the highest in-distribution F1 of 0.91, followed by the perplexity model (0.82). The commercial detector, not trained on Russian scientific text, reached 0.69. Paraphrased abstracts were the hardest category for all methods.

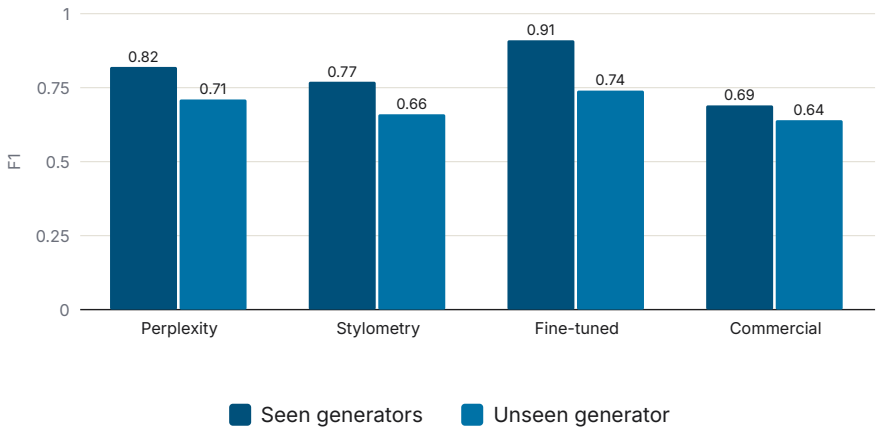


Fig. 1. Document-level F1 in-distribution and on an unseen generator

Generalisation to an unseen generator reduced performance substantially (Fig. 1). On a separate set of 400 human-written abstracts by authors for whom Russian is a second language, the fine-tuned classifier flagged 11% as machine-generated, compared with 3% for native-speaker abstracts.

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Table 1. Detection performance at document and sentence level

Method	F1 (doc)	F1 (sentence)	FPR, L2 authors
Perplexity features	0.82	0.61	0.14
Stylometric features	0.77	0.55	0.09
Fine-tuned transformer	0.91	0.68	0.11
Commercial detector	0.69	—	0.17

4 DISCUSSION

Our findings echo concerns raised for English: detectors that perform well on familiar generators degrade on new ones, and they disproportionately flag texts by non-native writers, whose prose tends to be lexically less varied. Given the rapid release of new models, in-distribution accuracy is a poor guide to real-world reliability.

For editorial practice, this implies that detector scores should at most prompt a conversation with authors, not trigger sanctions. Clear disclosure policies and attention to substantive scientific quality remain more robust safeguards.

5 CONCLUSION

We release a benchmark of 12,000 Russian scientific abstracts and show that current detectors, while useful, are neither robust to new generators nor fair to non-native authors. Automated detection should complement, not replace, editorial judgement.

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